

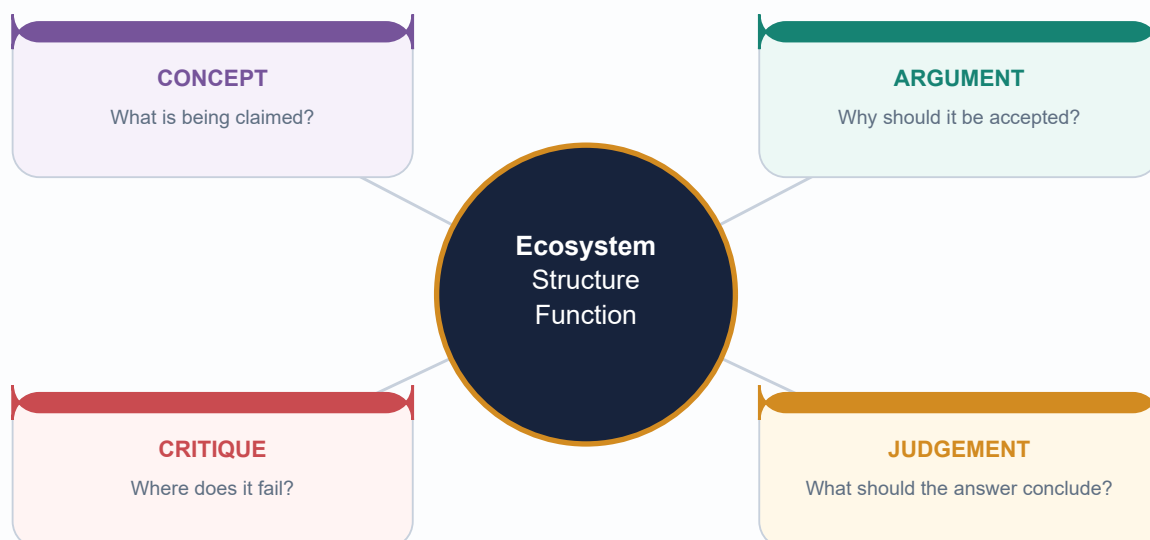
UPSC ENVIRONMENT AND ECOLOGY TOPIC PACKAGE

Ecosystem Structure and Function Complete Topic Package

Subject: Environment and Ecology | Paper: GS-III

Sources: 01_Ecosystem-Structure-and-Function.md + 01_Ecosystem-Structure-and-Function.md

Practice is derived from explicit chapter traps, not generic placeholder questions



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Foundation · Core Foundation

GS-III

Environment and Ecology

INTRO & ORIGIN

Subject: Environment and Ecology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III (Environment) + Prelims.

Foundation learning layer: concepts → constitutional detail → UPSC application.

CORE FOUNDATION: RECALL MAP

Core Foundation

V

Dimension 1

Subject: Environment and Ecology | Tier: Must-Do (foundation) | GS Paper: GS-III (Environment) + Prelims.

Dimension 2

Core area: Ecological fundamentals.

Dimension 3

Grounded in: NCERT Class XII Biology, Ch. "Ecosystem"; NIOS Environmental Studies; Majid Husain, *Indian and World Geography*, Ch. 5 "Biogeography"; aud

Dimension 4

□ = source-grounded | □□ = analytical linkage | □ = dated current-affairs anchor.

STATIC THEORY

- **Core area:** Ecological fundamentals.
- **Grounded in:** NCERT Class XII Biology, Ch. "Ecosystem"; NIOS Environmental Studies; Majid Husain, *Indian and World Geography*, Ch. 5 "Biogeography"; audited UPSC Environment PYQs (2024-2026 Prelims and 2024-2025 Mains).
- □ = source-grounded | □□ = analytical linkage | □ = dated current-affairs anchor.
- *Companion: advanced/01_Ecosystem-Structure-and-Function.md.*

MUST-KNOW FACTS

1. = source-grounded | □□ = analytical linkage | □ = dated current-affairs anchor.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → Core Foundation

HIGH

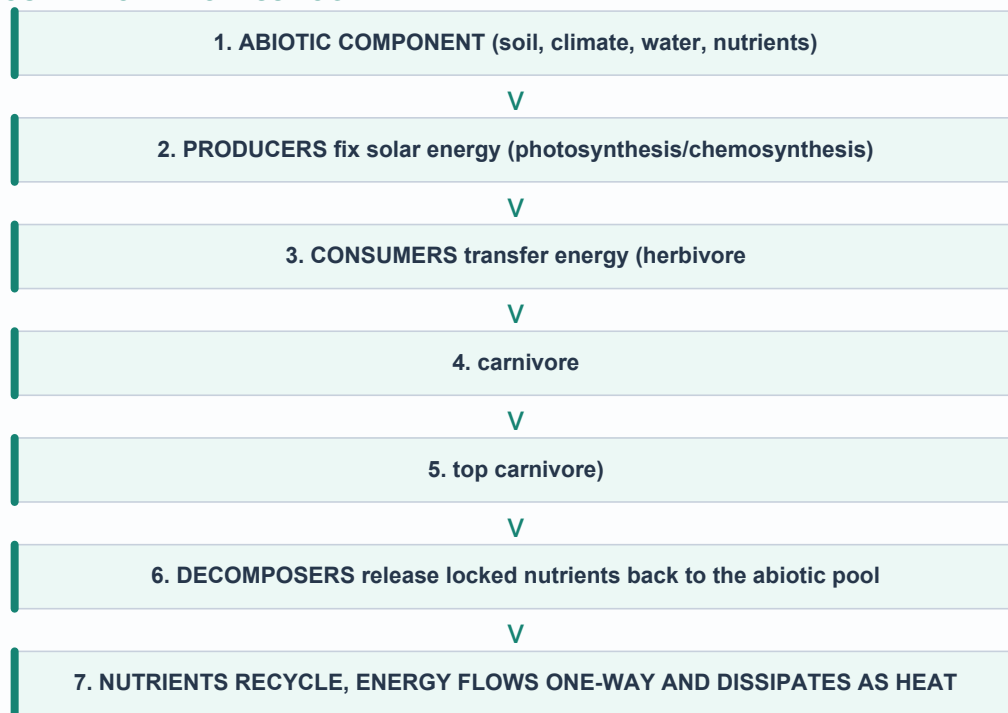
Foundation · Visual foundation · Visual 1

GS-III

Environment and Ecology

INTRO & ORIGIN

Visual-first reconstruction of the source logic.

VISUAL FOUNDATION: PROCESS LOGIC**MAINS ANGLE****STUDY LINK**

Ecosystem Structure and Function → Visual foundation

HIGH**Foundation · Visual foundation****GS-III**

Environment and Ecology

INTRO & ORIGIN

Core proposition: An ecosystem is a self-regulating functional unit in which energy flows unidirectionally while matter cycles; structure (who is present) and function (what they do) must both be read together to answer any ecosystem question.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Visual foundation

HIGH**Foundation · Essential definitions · Table 1.1****GS-III**

Environment and Ecology

INTRO & ORIGIN

Structured comparison from the foundation source layer.

KEY DATA

Concept	Exam-ready meaning
□ Ecosystem	A self-regulating functional unit of interacting biotic communities and their abiotic (physical and chemical) environment exchanging energy and matter. The term was coined by the British ecologist A.G. Tansley (1935) .
□ Biotic component	Living part — producers, consumers (herbivore/carnivore/omnivore), decomposers/detrivores. The biotic subsystems are conventionally read as plants, animals (including humans) and micro-organisms.
□ Abiotic component	Non-living part — climatic (temperature, light, rainfall) and edaphic (soil, minerals, topography) factors; at the biosphere scale this means the lithosphere, atmosphere and hydrosphere.
□ Habitat vs niche	Habitat is the physical address of a species; niche is its functional role, resource use and interactions — its "occupation" or "job" in the community.
□ Three components of a niche	A niche has a habitat niche (where it lives), a trophic/food niche (how it feeds) and a reproductive niche (how it breeds) — the standard biogeography decomposition.
□ Ecotone	Boundary transition zone between two adjoining ecosystems (e.g., mangrove, grassland-forest edge), varying in width and acting as a zone of *tension* where species of both communities compete for resources.
□ Edge effect	Tendency for greater species density/variety at the ecotone/community junction.
□ Biomass	Net dry weight of organic material — the material base that actually feeds the food chain.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Essential definitions

HIGH

Foundation · Essential definitions · Table 1.2

GS-III
Environment and
Ecology

INTRO & ORIGIN

Structured comparison from the foundation source layer.

KEY DATA

Concept	Exam-ready meaning
□ Life zone	Altitudinal zonation of plants and animals forming distinctive communities, each with its own temperature-precipitation regime.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Essential definitions

HIGH

Foundation · Topic mechanism · Part 1

GS-III
Environment and
Ecology

INTRO & ORIGIN

Producers (autotrophs) convert solar/chemical energy into chemical (food) energy — the

Foundation learning layer: concepts → constitutional detail → UPSC application.

TOPIC MECHANISM: RECALL MAP

Topic mechanism

V

Dimension 1 Producers (autotrophs) convert solar/chemical energy into chemical (food) energy — the	Dimension 2 only true energy entry point of an ecosystem.
Dimension 3 Energy moves through trophic levels via food chains; interlocking chains form a food web,	Dimension 4 which gives an ecosystem resilience against the loss of any single link.

STATIC THEORY

- only true energy entry point of an ecosystem.
- Energy moves through trophic levels via food chains; interlocking chains form a food web,
- which gives an ecosystem resilience against the loss of any single link.
- At each transfer, most energy is lost as metabolic heat (second law of thermodynamics), so
- energy flow is one-way and progressively diminishing — it is never recycled.
- Decomposers (bacteria, fungi) break down dead organic matter, releasing nutrients back into

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → Topic mechanism

HIGH

Foundation · Topic mechanism · Part 2

GS-III
Environment and
Ecology

INTRO & ORIGIN

soil/water/air, which producers reabsorb — this is why matter, unlike energy, cycles.

Foundation learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- Two structural components — stratification (vertical layering of species, e.g., canopy to forest floor) and productivity (rate of biomass generation) — define how "developed" an ecosystem is.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Topic mechanism

HIGH**Foundation · Institutions and policy tools · Part 1****GS-III**

Environment and Ecology

INTRO & ORIGIN

- **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention

Foundation learning layer: concepts → constitutional detail → UPSC application.

INSTITUTIONS AND POLICY TOOLS: RECALL MAP**Institutions and policy tools**

V

Dimension 1 □ MoEFCC: apex ministry for ecosystem/biodiversity policy and international convention	Dimension 2 implementation.
Dimension 3 □ Wildlife Institute of India (WII), Dehradun: research on ecosystem and species	Dimension 4 ecology supporting management plans.

STATIC THEORY

- implementation.
- □ **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
- ecology supporting management plans.
- □ **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and
- ecological surveys of flora and fauna underpinning ecosystem assessments.
- □ □ Ecosystem concepts underlie every protected-area, EIA and afforestation decision, so this

MUST-KNOW FACTS

1. **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention
2. **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
3. **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Institutions and policy tools

HIGH

Foundation · Institutions and policy tools · Part 2

GS-III
Environment and Ecology

INTRO & ORIGIN

topic is the base layer for Topics 04-16.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → Institutions and policy tools

HIGH

Foundation · Indian applications and examples

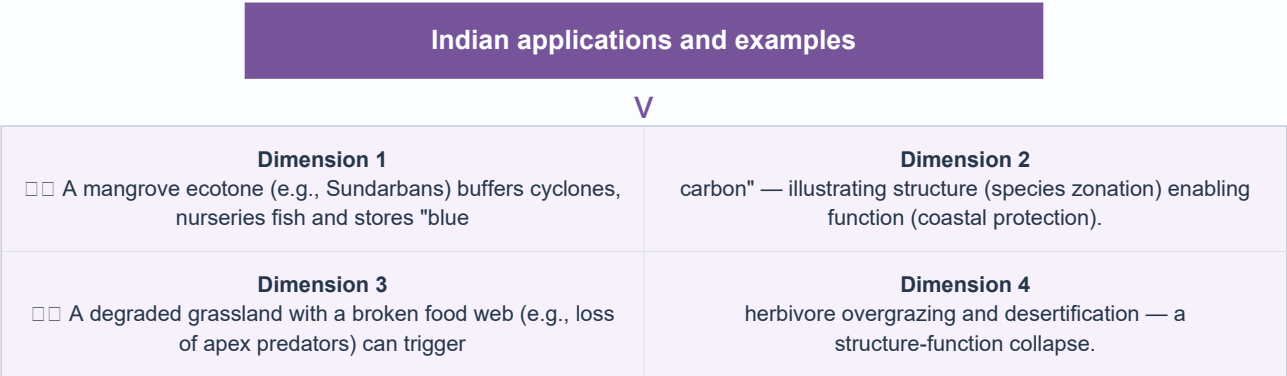
GS-III
Environment and Ecology

INTRO & ORIGIN

□□ A mangrove ecotone (e.g., Sundarbans) buffers cyclones, nurseries fish and stores "blue

Foundation learning layer: concepts → constitutional detail → UPSC application.

INDIAN APPLICATIONS AND EXAMPLES: RECALL MAP



STATIC THEORY

- carbon" — illustrating structure (species zonation) enabling function (coastal protection).
- □□ A degraded grassland with a broken food web (e.g., loss of apex predators) can trigger
- herbivore overgrazing and desertification — a structure-function collapse.
- □□ Western Ghats montane forest stratification (canopy, understorey, shrub, floor) supports
- high endemism because each layer offers a distinct niche.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → Indian applications and examples

HIGH

Foundation · Must-Know Facts for Prelims · Part 1

GS-III
Environment and
Ecology

INTRO & ORIGIN

- Producers are the sole entry point of energy into an ecosystem; energy flow is

Foundation learning layer: concepts → constitutional detail → UPSC application.

MUST-KNOW FACTS FOR PRELIMS: RECALL MAP

Must-Know Facts for Prelims

V

Dimension 1 □ Producers are the sole entry point of energy into an ecosystem; energy flow is	Dimension 2 unidirectional and matter/nutrient flow is cyclic.
Dimension 3 □ A food web is a network of interconnected food chains, giving greater ecosystem stability	Dimension 4 than a single linear chain.

STATIC THEORY

- unidirectional and matter/nutrient flow is cyclic.
- □ A food web is a network of interconnected food chains, giving greater ecosystem stability
- than a single linear chain.
- □ Decomposers are heterotrophic and act on dead organic matter; detritivores (e.g., earthworms) physically fragment it before microbial decomposition.
- □ Ecotones typically show the "edge effect" — higher species richness than either

MUST-KNOW FACTS

1. Producers are the sole entry point of energy into an ecosystem; energy flow is
2. A food web is a network of interconnected food chains, giving greater ecosystem stability
3. Decomposers are heterotrophic and act on dead organic matter; detritivores (e.g., earthworms) physically fragment it before microbial decomposition.
4. Ecotones typically show the "edge effect" — higher species richness than either

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Must-Know Facts for Prelims

HIGH

Foundation · Must-Know Facts for Prelims · Part 2

GS-III
Environment and
Ecology

INTRO & ORIGIN

neighbouring community.

Foundation learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- □ Niche cannot be shared identically by two species in the same habitat (competitive exclusion) — the basis of resource partitioning.
- □ The term "ecosystem" was coined by **A.G. Tansley (1935)**; the biosphere's abiotic frame is the lithosphere + atmosphere + hydrosphere.
- □ A niche is decomposed into habitat, trophic (food) and reproductive niches — a frequently examined three-part classification.

MUST-KNOW FACTS

1. Niche cannot be shared identically by two species in the same habitat (competitive exclusion)
2. The term "ecosystem" was coined by **A.G. Tansley (1935)**; the biosphere's abiotic frame is the lithosphere + atmosphere + hydrosphere.
3. A niche is decomposed into habitat, trophic (food) and reproductive niches — a frequently examined three-part classification.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Must-Know Facts for Prelims

HIGH

Foundation · Must-Know Facts for Prelims · Part 3

GS-III
Environment and
Ecology

INTRO & ORIGIN

- **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —

Foundation learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- the standard basis for pyramid-of-biomass comparisons.

MUST-KNOW FACTS

1. **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function →
Must-Know Facts for Prelims

HIGH**Foundation · UPSC traps****GS-III**Environment and
Ecology**INTRO & ORIGIN**

only matter/nutrients cycle.

Foundation learning layer: concepts → constitutional detail → UPSC application.

UPSC TRAPS: RECALL MAP**UPSC traps**

V

Dimension 1 only matter/nutrients cycle.	Dimension 2 role and resource use.
Dimension 3 beetles) are equally central to decomposition.	Dimension 4 interconnected food chains and gives more stability.

STATIC THEORY

- role and resource use.
- beetles) are equally central to decomposition.
- interconnected food chains and gives more stability.
- vents) are also producers without sunlight.
- ecological tension/competition, not a line on a map.
- weight** of organic material.

UPSC TRAPS

WRONG: Energy cycles like nutrients in an ecosystem.

CORRECT: Energy flows one-way and dissipates;

WRONG: Habitat and niche are the same thing.

CORRECT: Habitat is the address; niche is the functional

WRONG: Decomposers are only bacteria.

CORRECT: Fungi and detritivores (e.g., earthworms, dung

WRONG: A food chain and a food web are interchangeable terms.

CORRECT: A food web is the sum of

WRONG: Producers are always green plants.

CORRECT: Chemosynthetic bacteria (e.g., at hydrothermal

WRONG: An ecotone is simply a boundary line.

CORRECT: It is a *zone* of variable width and a zone of

WRONG: "Biomass" means the live weight of organisms.

CORRECT: It is conventionally the ****net dry**

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → UPSC traps

HIGH**Foundation · □ Current anchor**

GS-III
Environment and
Ecology

NEWS TRIGGER

□ As of the 18th India State of Forest Report (ISFR 2023, released December 2024 — the latest ISFR edition verified for this knowledge base as on **2 August 2026**), India's total forest and tree cover is 8,27,357 sq km — 25.17% of the geographical area (forest cover 21.76%, tree cover 3.41%) — the base ecosystem-extent statistic to cite with its date.

□□ ISFR is a **biennial** FSI publication, so verify against the newest edition on fsi.nic.in before quoting in an answer written later than this date. □ Economic Survey 2025-26 (Ch. 10, *Environment and Climate Change*) frames India's

INTRO & ORIGIN

policy direction as "ecosystem-led, development-integrated climate resilience" — the government's own vocabulary for treating ecosystem function (mangroves, wetlands, forests) as adaptation infrastructure rather than as a separate conservation silo.

Foundation learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- □□ **Interpretation caution:** forest/tree cover statistics measure canopy extent, not ecosystem health, species diversity or forest quality — do not conflate the two in answers.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → □
Current anchor

HIGH**Foundation · PYQ application****GS-III**Environment and
Ecology**INTRO & ORIGIN**

Exam-ready treatment of PYQ application.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

□□ *Recurring Prelims pattern: identify correct statements on energy flow, trophic levels and decomposer function using elimination against the traps above.* □□ *Mains linkage: ecosystem-services framing (provisioning, regulating, supporting, cultural) is used to justify conservation and EIA arguments across GS-III.*

STUDY LINK

Ecosystem Structure and Function → PYQ
application

HIGH**Foundation · Mains angles****GS-III**Environment and
Ecology**INTRO & ORIGIN**

Exam-ready treatment of Mains angles.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

□□ *Frame any ecosystem answer around structure (components, stratification) enabling function (energy flow, nutrient cycling, resilience).*
□□ *Link ecosystem degradation to loss of ecosystem services and downstream socio-economic costs (e.g., loss of pollination, flood buffering).* □□ *Conclude with an ecosystem-based-management thesis: policy must protect structure to preserve function, not just protect an isolated flagship species. **Answer thesis:** Treat structure (components and their arrangement) and function (energy flow, matter cycling, resilience) as two faces of the same ecosystem, and evaluate any disturbance by asking whether it breaks a structural link that a critical function depends on.*

STUDY LINK

Ecosystem Structure and Function → Mains
angles

HIGH

Foundation · Probable questions

GS-III

Environment and
Ecology

INTRO & ORIGIN

□□ **Prelims:** Distinguish habitat, niche, ecotone and edge effect with an Indian example.

Foundation learning layer: concepts → constitutional detail → UPSC application.

PROBABLE QUESTIONS: RECALL MAP

Probable questions

V

Dimension 1

□□ Prelims: Distinguish habitat, niche, ecotone and edge effect with an Indian example.

Dimension 2

□□ Mains (10 marks): Explain why energy flow in an ecosystem is unidirectional while

Dimension 3

nutrient flow is cyclic.

Dimension 4

□□ Mains (15 marks): Discuss how loss of stratification/keystone species destabilises

STATIC THEORY

- □□ **Mains (10 marks):** Explain why energy flow in an ecosystem is unidirectional while
- nutrient flow is cyclic.
- □□ **Mains (15 marks):** Discuss how loss of stratification/keystone species destabilises
- ecosystem function, with a Western Ghats or Sundarbans example.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Probable questions

HIGH

Foundation · Study links

GS-III

Environment and
Ecology

INTRO & ORIGIN

□ Advanced companion: advanced/01_Ecosystem-Structure-and-Function.md.

Foundation learning layer: concepts → constitutional detail → UPSC application.

STUDY LINKS: RECALL MAP

Study links

V

Dimension 1

□ Advanced companion:
advanced/01Ecosystem-Structure-and-Function.md.

Dimension 2

□ 02Biogeochemical-Cycles-and-Ecological-Pyramids.md —
quantifying the energy/matter

Dimension 3

flows introduced here.

Dimension 4

□ 03Ecological-Succession-and-Biomes.md — how
ecosystems develop and are classified

STATIC THEORY

- □ 02_Biogeochemical-Cycles-and-Ecological-Pyramids.md — quantifying the energy/matter flows introduced here.
- □ 03_Ecological-Succession-and-Biomes.md — how ecosystems develop and are classified globally.
- □ 04_Biodiversity-Levels-and-Hotspots.md — species-level consequence of ecosystem structure.

MUST-KNOW FACTS

1. Advanced companion: advanced/01_Ecosystem-Structure-and-Function.md.
2. 02_Biogeochemical-Cycles-and-Ecological-Pyramids.md — quantifying the energy/matter
3. 03_Ecological-Succession-and-Biomes.md — how ecosystems develop and are classified
4. 04_Biodiversity-Levels-and-Hotspots.md — species-level consequence of ecosystem

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Study links

HIGH**Advanced · Core Foundation****GS-III**

Environment and Ecology

INTRO & ORIGIN

Subject: Environment and Ecology | **Tier:** Advanced | **GS Paper:** GS-III (Environment) + Prelims.

Advanced learning layer: concepts → constitutional detail → UPSC application.

CORE FOUNDATION: RECALL MAP**Core Foundation**

V

Dimension 1

Subject: Environment and Ecology | Tier: Advanced | GS Paper: GS-III (Environment) + Prelims.

Dimension 2

Core area: Ecological fundamentals.

Dimension 3

Grounded in: NCERT Class XII Biology, Ch. "Ecosystem"; MoEFCC/ISFR data; Economic Survey 2025-26 Ch. 10 (Environment and Climate Change); ecosystem-se

Dimension 4

□ = source-grounded | □□ = inference/analysis | □ = dated current-affairs anchor.

STATIC THEORY

- **Core area:** Ecological fundamentals.

- **Grounded in:** NCERT Class XII Biology, Ch. "Ecosystem"; MoEFCC/ISFR data; Economic Survey 2025-26 Ch. 10 (Environment and Climate Change); ecosystem-services literature (MEA framework); audited UPSC Environment PYQs (2024-2026 Prelims and 2024-2025 Mains).

- □ = source-grounded | □□ = inference/analysis | □ = dated current-affairs anchor.

- *Companion: basic/01_Ecosystem-Structure-and-Function.md.*

MUST-KNOW FACTS

1. = source-grounded | □□ = inference/analysis | □ = dated current-affairs anchor.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Core Foundation

HIGH**Advanced · Why this is the base layer, not just "Biology revision"**

GS-III
Environment and
Ecology

INTRO & ORIGIN

□□ Every downstream Environment topic in this knowledge base — biodiversity loss, protected areas, EIA, forest law, climate impacts — is ultimately an argument about a structural component of an ecosystem being altered and a function it performed being lost or degraded. Examiners test whether a candidate can trace that causal chain, not whether they can define "ecosystem."

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Why this is the base layer, not just "Biology revision"

HIGH**Advanced · The ecosystem-services lens (analytical upgrade)**

GS-III
Environment and
Ecology

INTRO & ORIGIN

Structured comparison from the advanced source layer.

KEY DATA

Service category	Definition	Loss example if degraded
□□ Provisioning	Direct material outputs — food, timber, water, fibre.	Fishery collapse from wetland loss.
□□ Regulating	Ecosystem control of natural processes — flood buffering, pollination, climate moderation.	Mangrove removal raising cyclone-surge damage.
□□ Supporting	Underlying processes enabling all other services — nutrient cycling, soil formation, primary production.	Soil biota loss reducing long-term fertility.
□□ Cultural	Non-material benefits — recreation, spiritual, aesthetic, scientific value.	Loss of sacred groves reducing community-led conservation incentive.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → The ecosystem-services lens (analytical upgrade)

HIGH

Advanced · The ecosystem-services lens (analytical upgrade)

GS-III
Environment and
Ecology

INTRO & ORIGIN

□□ This four-category framework (drawn from the globally used Millennium Ecosystem Assessment structure) is the standard vocabulary examiners expect when a question asks candidates to "assess the value of an ecosystem" rather than merely describe it.

Advanced learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- □□ **Analytical upgrade — MEA is not the end of the story.** The MEA (2005) classification has since been reframed by IPBES (the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services) as "**Nature's Contributions to People**" (NCP), which deliberately widens the lens beyond a market/utilitarian reading to include relational and indigenous/local-knowledge values of nature. A high-scoring answer names MEA for the four-fold taxonomy but flags the NCP reframing as the reason "ecosystem services" is no longer treated as a purely economic-valuation concept.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → The ecosystem-services lens (analytical upgrade)

HIGH

Advanced · 2A. Ecosystem function as adaptation infrastructure (the current policy frame)

GS-III
Environment and
Ecology

INTRO & ORIGIN

Structured comparison from the advanced source layer.

KEY DATA

Instrument	Ecosystem structure protected	Function purchased	Verified anchor
□ MISHTI (Mangrove Initiative for Shoreline Habitats & Tangible Incomes)	Mangrove zonation along the coast	Storm-surge buffering, fish nursery, blue carbon	~540 sq km restoration across 9 coastal States and 4 UTs , 2023-2028; ~22.8 million person-days of employment; estimated carbon sink ~4.5 million tonnes (Economic Survey 2025-26, Ch. 10)
□ National Plan for Conservation of Aquatic Ecosystems (NPCA)	Wetland hydrology and vegetation	Flood moderation, water security	ES 2025-26 credits NPCA with expanding protected wetlands that act as "critical buffers under climate stress"
□ National Coastal Mission	Coastal landform-vegetation complex	Reduced vulnerability to sea-level rise/extreme events	ES 2025-26, Ch. 10

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → 2A.
Ecosystem function as adaptation infrastructure (the current policy frame)

HIGH

Advanced · 2A. Ecosystem function as adaptation infrastructure (the current policy frame)

GS-III
Environment and
Ecology

NEWS TRIGGER

□□ The operational significance of "structure enables function" in Indian policy today is that ecosystem function is being treated as **infrastructure**, not sentiment: □□ **Why this matters for marking:** it converts an abstract "ecosystem services" paragraph into a named-scheme, dated, quantified argument — the difference between a 5/10 and an 8/10 answer on ecosystem-value questions.

INTRO & ORIGIN

Exam-ready treatment of 2A. Ecosystem function as adaptation infrastructure (the current policy frame).

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → 2A.
Ecosystem function as adaptation infrastructure (the current policy frame)

HIGH

Advanced · Structural analysis: stratification, productivity, and stability

GS-III
Environment and
Ecology

INTRO & ORIGIN

Stratification (vertical species layering) is a proxy for niche differentiation — more

Advanced learning layer: concepts → constitutional detail → UPSC application.

STRUCTURAL ANALYSIS: STRATIFICATION, PRODUCTIVITY, AND STABILITY: RECALL MAP

Structural analysis: stratification, productivity, and stability

V

Dimension 1 Stratification (vertical species layering) is a proxy for niche differentiation — more	Dimension 2 layers generally mean more niches and hence more species can coexist without direct competition (resource partitioning).
Dimension 3 Productivity — gross primary productivity (GPP) minus respiration gives net primary	Dimension 4 productivity (NPP), the biomass actually available to the next trophic level; only 10% or so of a level's energy typically reaches the next (the eco

STATIC THEORY

- layers generally mean more niches and hence more species can coexist without direct competition (resource partitioning).
- **Productivity** — gross primary productivity (GPP) minus respiration gives net primary
- productivity (NPP), the biomass actually available to the next trophic level; only 10% or so of a level's energy typically reaches the next (the ecological efficiency argument used to justify why food chains rarely exceed 4-5 links).
- **Stability vs resilience:** a "stable" ecosystem resists change; a "resilient" one
- recovers after disturbance. □ □ UPSC answers should not conflate the two — a monoculture plantation can look structurally uniform (stable-looking) yet be ecologically fragile (low resilience to pest outbreaks or drought).

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Structural analysis: stratification, productivity,
and stability

HIGH

Advanced · Governance and implementation linkage

GS-III
Environment and
Ecology

INTRO & ORIGIN

□ **MoEFCC** operationalises ecosystem concepts through species/habitat law (Topics 06-10),

Advanced learning layer: concepts → constitutional detail → UPSC application.

GOVERNANCE AND IMPLEMENTATION LINKAGE: RECALL MAP

Governance and implementation linkage

V

Dimension 1 ☐ MoEFCC operationalises ecosystem concepts through species/habitat law (Topics 06-10),	Dimension 2 forest law (Topics 11-12) and clearance regimes (Topic 16).
Dimension 3 ☐ WII, BSI, ZSI generate the baseline ecological data (species inventories, habitat	Dimension 4 assessments) on which protected-area boundaries and EIA studies are built.

STATIC THEORY

- forest law (Topics 11-12) and clearance regimes (Topic 16).
- ☐ **WII, BSI, ZSI** generate the baseline ecological data (species inventories, habitat
- assessments) on which protected-area boundaries and EIA studies are built.
- ☐ ☐ **Implementation gap:** ecosystem-services value is rarely monetised in Indian project
- appraisal, so an EIA can approve a project after assessing only a narrow set of listed environmental parameters while ignoring harder-to-quantify supporting/cultural services — a recurring critique in Mains answers on EIA effectiveness (cross-refer Topic 16).

MUST-KNOW FACTS

1. **MoEFCC** operationalises ecosystem concepts through species/habitat law (Topics 06-10),
2. **WII, BSI, ZSI** generate the baseline ecological data (species inventories, habitat

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Governance and implementation linkage

HIGH

Advanced · Data and conceptual limitations
(answer-writing caution)

GS-III
Environment and
Ecology

INTRO & ORIGIN

Exam-ready treatment of Data and conceptual limitations (answer-writing caution).

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

☐ ☐ *Forest/tree cover statistics (ISFR) measure canopy density thresholds, not species composition, age structure or ecosystem function — a monoculture plantation and an old- growth forest can register identically as "forest cover."* ☐ ☐ *"Ecosystem health" has no single agreed metric; examiners reward candidates who name specific indicators (species richness, NPP, nutrient-cycling rate, resilience to disturbance) rather than asserting health in the abstract.* ☐ ☐ *Keystone-species arguments (e.g., apex predator removal triggering trophic cascade) are well-evidenced globally but site-specific data for many Indian ecosystems remains thin — qualify such claims as illustrative rather than universally quantified.*

STUDY LINK

Ecosystem Structure and Function → Data and conceptual limitations (answer-writing caution)

HIGH**Advanced · Recurring UPSC analytical tensions****GS-III**
Environment and
Ecology**INTRO & ORIGIN**

Structured comparison from the advanced source layer.

KEY DATA

Tension	Balanced framing
Single-species conservation vs ecosystem conservation	A flagship species (tiger, elephant) can be an "umbrella" that indirectly protects whole ecosystem structure — but only if habitat connectivity, not just the animal, is protected.
Productivity vs biodiversity	High-productivity systems (monocultures) are not necessarily high-biodiversity or high-resilience systems.
Development vs ecosystem-service loss	Growth needs land/water; the ecosystem-services framework lets planners price the foregone regulating/supporting services rather than treat them as free.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function →
Recurring UPSC analytical tensions

HIGH**Advanced · Must-Know Facts for Advanced Prelims · Part 1****GS-III**
Environment and
Ecology**INTRO & ORIGIN**

- Only about 10% of energy is transferred on average from one trophic level to the next

*Advanced learning layer: concepts → constitutional detail → UPSC application.***MUST-KNOW FACTS FOR ADVANCED PRELIMS: RECALL MAP****Must-Know Facts for Advanced Prelims****V**

Dimension 1 □ Only about 10% of energy is transferred on average from one trophic level to the next	Dimension 2 (ecological efficiency), which is why food chains are typically short.
Dimension 3 □ Net Primary Productivity (NPP) = Gross Primary Productivity (GPP) – respiratory loss by	Dimension 4 producers; NPP is what is available to consumers.

STATIC THEORY

- (ecological efficiency), which is why food chains are typically short.
- □ Net Primary Productivity (NPP) = Gross Primary Productivity (GPP) – respiratory loss by producers; NPP is what is available to consumers.
- □ The Millennium Ecosystem Assessment's four-fold service classification (provisioning, regulating, supporting, cultural) is the standard analytical vocabulary for ecosystem-value questions.
- □ Ecotones show the edge effect — higher species density and variety than either bordering

MUST-KNOW FACTS

1. Only about 10% of energy is transferred on average from one trophic level to the next
2. Net Primary Productivity (NPP) = Gross Primary Productivity (GPP) – respiratory loss by producers
3. The Millennium Ecosystem Assessment's four-fold service classification (provisioning, regulating, supporting, cultural) is the standard analytical vocabulary for ecosystem-value questions.
4. Ecotones show the edge effect — higher species density and variety than either bordering

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function →
Must-Know Facts for Advanced Prelims

HIGH**Advanced · Must-Know Facts for Advanced Prelims · Part 2**

GS-III
Environment and
Ecology

INTRO & ORIGIN

community due to overlapping resource availability.

Advanced learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- □ IPBES reframed the MEA's "ecosystem services" as **Nature's Contributions to People
- (NCP)** to admit relational and indigenous/local-knowledge values alongside instrumental ones — the vocabulary shift examiners increasingly reward.
- □ A niche is not a single attribute: it decomposes into habitat, trophic (food) and reproductive niches, which is why two species can overlap in one dimension and still coexist.

MUST-KNOW FACTS

1. IPBES reframed the MEA's "ecosystem services" as **Nature's Contributions to People**
2. A niche is not a single attribute: it decomposes into habitat, trophic (food) and

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function →
Must-Know Facts for Advanced Prelims

HIGH**Advanced · Advanced Prelims traps****GS-III**Environment and
Ecology**INTRO & ORIGIN**

change) and resilience (capacity to recover) are distinct properties.

Advanced learning layer: concepts → constitutional detail → UPSC application.

ADVANCED PRELIMS TRAPS: RECALL MAP**Advanced Prelims traps**

V

Dimension 1 change) and resilience (capacity to recover) are distinct properties.	Dimension 2 not composition or ecological function.
Dimension 3 only if habitat structure and connectivity are also protected (umbrella-species logic).	Dimension 4 heat/metabolism at each transfer.

STATIC THEORY

- not composition or ecological function.
- only if habitat structure and connectivity are also protected (umbrella-species logic).
- heat/metabolism at each transfer.

UPSC TRAPS

WRONG: A stable-looking ecosystem is automatically resilient.

CORRECT: Stability (resistance to

WRONG: Forest cover data reflects ecosystem quality.

CORRECT: ISFR measures canopy density thresholds,

WRONG: Protecting one flagship species is equivalent to protecting the ecosystem.

CORRECT: It works

WRONG: Energy efficiency between trophic levels is close to 100%.

CORRECT: Roughly 90% is lost as

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
Advanced Prelims traps

HIGH

Advanced · ☐ Current anchor — analytical use

GS-III
Environment and
Ecology

INTRO & ORIGIN

Structured comparison from the advanced source layer.

KEY DATA

Verified current anchor

Topic-specific analytical use

☐ ISFR 2023 (December 2024; latest ISFR verified as on 2 August 2026): forest+tree cover 8,27,357 sq km, 25.17% of geographical area (forest 21.76%, tree cover 3.41%).

Use as the quantitative base for any "extent of forest ecosystem" answer, but immediately flag the quality/quantity distinction to earn analytical credit. ISFR is biennial — always re-verify the edition.

☐ Economic Survey 2025-26 (Ch. 10): India's adaptation-relevant expenditure rose to **5.6% of GDP in FY22** (source: India's Initial Adaptation Communication / Third National Communication to UNFCCC, 2023).

Converts "ecosystems are valuable" into a fiscal claim: India is already spending at ecosystem-adaptation scale, so the policy question is **allocative efficiency**, not whether to spend.

☐ Economic Survey 2025-26 (Ch. 10): MISHTI targets ~540 sq km of mangrove restoration across 9 coastal States and 4 UTs (2023-2028), with an estimated 4.5 million tonne carbon sink.

The single best worked example of "structure (mangrove zonation) purchased for function (surge buffering + blue carbon + livelihoods)".

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → ☐
Current anchor — analytical use

HIGH

Advanced · ☐ Current anchor — analytical use

GS-III
Environment and
Ecology

NEWS TRIGGER

☐☐ **Announced-target vs achievement discipline:** MISHTI's 540 sq km and 4.5 Mt figures are *programme targets/estimates*, not audited outcomes — say "targets/envisages", never "has restored", unless a completion figure is cited.

INTRO & ORIGIN

Exam-ready treatment of ☐ Current anchor — analytical use.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function → ☐
Current anchor — analytical use

HIGH

Advanced · PYQ-based analytical application

GS-III
Environment and
Ecology

INTRO & ORIGIN

Exam-ready treatment of PYQ-based analytical application.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

□□ *Trophic-level and energy-flow statement-based Prelims questions are answered fastest by applying the one-way-energy / cyclic-matter rule and the ecological-efficiency figure.* □□ *Mains questions on "ecosystem services" or "valuing nature" expect the four-category framework plus a governance-gap critique (Section 4-5 above), not a textbook definition.*

STUDY LINK

Ecosystem Structure and Function →
PYQ-based analytical application

HIGH

Advanced · Mains-ready framework

GS-III
Environment and
Ecology

INTRO & ORIGIN

Exam-ready treatment of Mains-ready framework.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

Central thesis: *Treat structure and function as inseparable; evaluate any disturbance, policy or project by asking which structural component it removes and which ecosystem service consequently fails — then assess whether current governance (EIA, forest law, protected-area law) actually prices or protects that service. Define ecosystem structure (biotic + abiotic components, stratification) and function (energy flow, nutrient cycling, resilience). Introduce the four-fold ecosystem-services framework as the analytical bridge to policy. Apply governance linkage — cite the institution (MoEFCC/WII/BSI/ZSI) and the legal instrument (EIA, forest law, protected-area law) relevant to the specific service at risk. State the implementation/data limitation (e.g., EIA rarely monetises supporting/cultural services; forest-cover metrics do not capture quality). Conclude with an ecosystem-based-management recommendation, not a single-species fix.*

STUDY LINK

Ecosystem Structure and Function →
Mains-ready framework

HIGH

Advanced · Probable questions

GS-III
Environment and
Ecology

INTRO & ORIGIN

□□ **Prelims:** Apply ecological-efficiency and one-way-energy-flow logic to eliminate

Advanced learning layer: concepts → constitutional detail → UPSC application.

PROBABLE QUESTIONS: RECALL MAP

Probable questions

V

Dimension 1 ☐ Prelims: Apply ecological-efficiency and one-way-energy-flow logic to eliminate	Dimension 2 incorrect statements on trophic transfer.
Dimension 3 ☐ Mains (10 marks): Explain the Millennium Ecosystem Assessment's four categories of	Dimension 4 ecosystem services with one Indian example each.

STATIC THEORY

- incorrect statements on trophic transfer.
- ☐ **Mains (10 marks):** Explain the Millennium Ecosystem Assessment's four categories of
- ecosystem services with one Indian example each.
- ☐ **Mains (15 marks):** "Protecting a flagship species is not the same as protecting an
- ecosystem." Critically examine with reference to Indian conservation practice.

MAINS ANGLE

STUDY LINK

Ecosystem Structure and Function →
 Probable questions

HIGH

Advanced · Study links

GS-III
 Environment and
 Ecology

INTRO & ORIGIN

- ☐ Foundation companion: basic/01_Ecosystem-Structure-and-Function.md.

Advanced learning layer: concepts → constitutional detail → UPSC application.

STUDY LINKS: RECALL MAP

Study links

V

Dimension 1 ☐ Foundation companion: basic/01Ecosystem-Structure-and-Function.md.	Dimension 2 ☐ 02Biogeochemical-Cycles-and-Ecological-Pyramids.md — quantitative treatment of the
Dimension 3 energy/matter flows analysed here.	Dimension 4 ☐ 16Environmental-Impact-Assessment-and-NGT.md — where the ecosystem-services pricing

STATIC THEORY

- □ 02_Biogeochemical-Cycles-and-Ecological-Pyramids.md — quantitative treatment of the energy/matter flows analysed here.
- □ 16_Environmental-Impact-Assessment-and-NGT.md — where the ecosystem-services pricing gap becomes a governance and litigation issue.
- □ 27_Environmental-Institutions-MoEFCC-CPCB-NBA-WII.md — mandates of the institutions
- cited in Section 4.

MUST-KNOW FACTS

1. Foundation companion: basic/01_Ecosystem-Structure-and-Function.md.
2. 02_Biogeochemical-Cycles-and-Ecological-Pyramids.md — quantitative treatment of the
3. 16_Environmental-Impact-Assessment-and-NGT.md — where the ecosystem-services pricing
4. 27_Environmental-Institutions-MoEFCC-CPCB-NBA-WII.md — mandates of the institutions

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function → Study links

HIGH**Current-Affairs Verification Note****GS-III**

Environment and Ecology

INTRO & ORIGIN

A live February-August 2026 scan was performed. No unverified headline is inserted. The source chapter's dated current anchor is retained where it is already verified.

STATIC THEORY

- Static concepts remain the evidentiary base.
- Current affairs are used only when the conceptual linkage is direct.
- No-current-item is preferable to a fabricated or weakly sourced linkage.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function -> Current application

HIGH**Topic-Specific MCQ Loop 1****GS-III**
Environment and
Ecology**INTRO & ORIGIN**

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q1. Which statement correctly resolves this common misconception: "Energy cycles like nutrients in an ecosystem."?

- A. Energy flows one-way and dissipates;
- B. Energy cycles like nutrients in an ecosystem.
- C. Habitat and niche are the same thing.
- D. Decomposers are only bacteria.

- ANSWER: A - Energy flows one-way and dissipates;

- Q2. Which statement correctly resolves this common misconception: "Habitat and niche are the same thing."?

- A. Habitat and niche are the same thing.
- B. Habitat is the address; niche is the functional
- C. Decomposers are only bacteria.
- D. A food chain and a food web are interchangeable terms.

- ANSWER: B - Habitat is the address; niche is the functional

- Q3. Which statement correctly resolves this common misconception: "Decomposers are only bacteria."?

- A. Decomposers are only bacteria.
- B. A food chain and a food web are interchangeable terms.
- C. Fungi and detritivores (e.g., earthworms, dung
- D. Producers are always green plants.

- ANSWER: C - Fungi and detritivores (e.g., earthworms, dung

- Q4. Which statement correctly resolves this common misconception: "A food chain and a food web are interchangeable terms."?

- A. A food chain and a food web are interchangeable terms.
- B. Producers are always green plants.
- C. An ecotone is simply a boundary line.
- D. A food web is the sum of

- ANSWER: D - A food web is the sum of

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function ->
Source-grounded practice

HIGH**Topic-Specific MCQ Loop 2****GS-III**Environment and
Ecology**INTRO & ORIGIN**

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q5. Which statement correctly resolves this common misconception: "Producers are always green plants."?
- A. Chemosynthetic bacteria (e.g., at hydrothermal
- B. Producers are always green plants.
- C. An ecotone is simply a boundary line.
- D. "Biomass" means the live weight of organisms.
- ANSWER: A - Chemosynthetic bacteria (e.g., at hydrothermal
- Q6. Which statement correctly resolves this common misconception: "An ecotone is simply a boundary line."?
- A. An ecotone is simply a boundary line.
- B. It is a *zone* of variable width and a zone of
- C. "Biomass" means the live weight of organisms.
- D. A stable-looking ecosystem is automatically resilient.
- ANSWER: B - It is a *zone* of variable width and a zone of
- Q7. Which statement correctly resolves this common misconception: "'Biomass' means the live weight of organisms."?
- A. "Biomass" means the live weight of organisms.
- B. A stable-looking ecosystem is automatically resilient.
- C. It is conventionally the **net dry**
- D. Energy cycles like nutrients in an ecosystem.
- ANSWER: C - It is conventionally the **net dry**
- Q8. Which statement correctly resolves this common misconception: "A stable-looking ecosystem is automatically resilient."?
- A. A stable-looking ecosystem is automatically resilient.
- B. Energy cycles like nutrients in an ecosystem.
- C. Habitat and niche are the same thing.
- D. Stability (resistance to
- ANSWER: D - Stability (resistance to

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**Ecosystem Structure and Function ->
Source-grounded practice**HIGH****Solved 10-Mark Practice****GS-III**
Environment and
Ecology**INTRO & ORIGIN**

Explain Ecosystem Structure and Function with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- ☐ **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention
- ☐ **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
- ☐ **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and
- ☐ ☐ Ecosystem concepts underlie every protected-area, EIA and afforestation decision, so this
- ☐ ☐ A mangrove ecotone (e.g., Sundarbans) buffers cyclones, nurseries fish and stores "blue
- ☐ ☐ A degraded grassland with a broken food web (e.g., loss of apex predators) can trigger
- ☐ ☐ Western Ghats montane forest stratification (canopy, understorey, shrub, floor) supports
- ☐ Producers are the sole entry point of energy into an ecosystem; energy flow is
- ☐ A food web is a network of interconnected food chains, giving greater ecosystem stability
- ☐ Decomposers are heterotrophic and act on dead organic matter; detritivores (e.g.,
- ☐ Ecotones typically show the "edge effect" — higher species richness than either
- ☐ Niche cannot be shared identically by two species in the same habitat (competitive
- ☐ The term "ecosystem" was coined by **A.G. Tansley (1935)**; the biosphere's abiotic
- ☐ A niche is decomposed into habitat, trophic (food) and reproductive niches — a
- ☐ **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

Ecosystem Structure and Function -> 10-mark answer

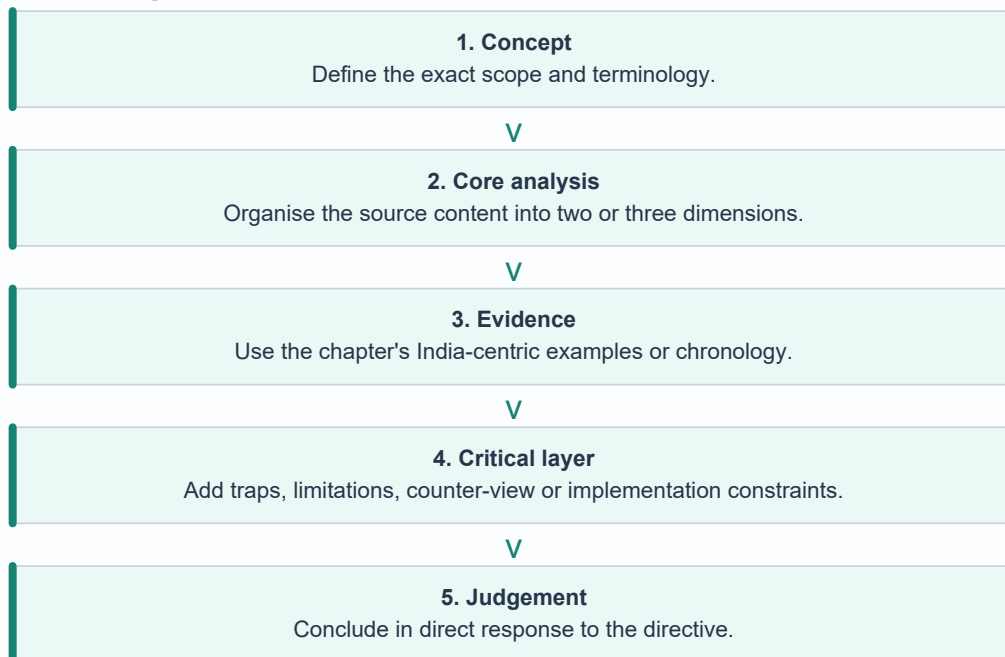
HIGH

Solved 15-Mark Practice

GS-III
Environment and
Ecology

INTRO & ORIGIN

Analyse Ecosystem Structure and Function with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**STATIC THEORY**

- ☐ **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention
- ☐ **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
- ☐ **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and
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MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

Ecosystem Structure and Function -> 15-mark answer

HIGH

Solved 20-Mark Practice

GS-III
Environment and
Ecology

INTRO & ORIGIN

Critically examine Ecosystem Structure and Function with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE

1. Concept

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- ☐ **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention
- ☐ **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
- ☐ **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and
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- ☐ A niche is decomposed into habitat, trophic (food) and reproductive niches — a
- ☐ **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

Ecosystem Structure and Function -> 20-mark answer

HIGH**Final Consolidated Register Notes**

GS-III
Environment and
Ecology

INTRO & ORIGIN

Final revision is placed after teaching and solved practice.

TOPIC RECALL MAP

Ecosystem Structure and Function

V

Definition ☐ MoEFCC : apex ministry for ecosystem/biodiversity policy and international convention	Mechanism ☐ Wildlife Institute of India (WII), Dehradun : research on ecosystem and species
Application ☐ Botanical Survey of India (BSI) / Zoological Survey of India (ZSI) : taxonomic and	Evaluation ☐ ☐ Ecosystem concepts underlie every protected-area, EIA and afforestation decision, so this

STATIC THEORY

- ☐ **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention
- ☐ **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
- ☐ **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and
- ☐ ☐ Ecosystem concepts underlie every protected-area, EIA and afforestation decision, so this
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- ☐ Ecotones typically show the "edge effect" — higher species richness than either
- ☐ Niche cannot be shared identically by two species in the same habitat (competitive

MEMORY HOOK

Meaning -> mechanism -> evidence -> limitation -> judgement

MUST-KNOW FACTS

1. □ The term "ecosystem" was coined by **A.G. Tansley (1935)**; the biosphere's abiotic
2. □ A niche is decomposed into habitat, trophic (food) and reproductive niches — a
3. □ **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —
4. □ As of the 18th India State of Forest Report (ISFR 2023, released December 2024 — the
5. □ Economic Survey 2025-26 (Ch. 10, *Environment and Climate Change*) frames India's
6. □□ Recurring Prelims pattern: identify correct statements on energy flow, trophic levels

UPSC TRAPS

- WRONG:** Energy cycles like nutrients in an ecosystem.
CORRECT: Energy flows one-way and dissipates;
- WRONG:** Habitat and niche are the same thing.
CORRECT: Habitat is the address; niche is the functional
- WRONG:** Decomposers are only bacteria.
CORRECT: Fungi and detritivores (e.g., earthworms, dung
- WRONG:** A food chain and a food web are interchangeable terms.
CORRECT: A food web is the sum of

MAINS ANGLE

Use Ecosystem Structure and Function as an argument, not as an unstructured list of facts.

STUDY LINK

Environment and Ecology -> Ecosystem
Structure and Function

End of Register Notes - UPSC Agent / Copilot CLI